

How MIAMA links active travel to health

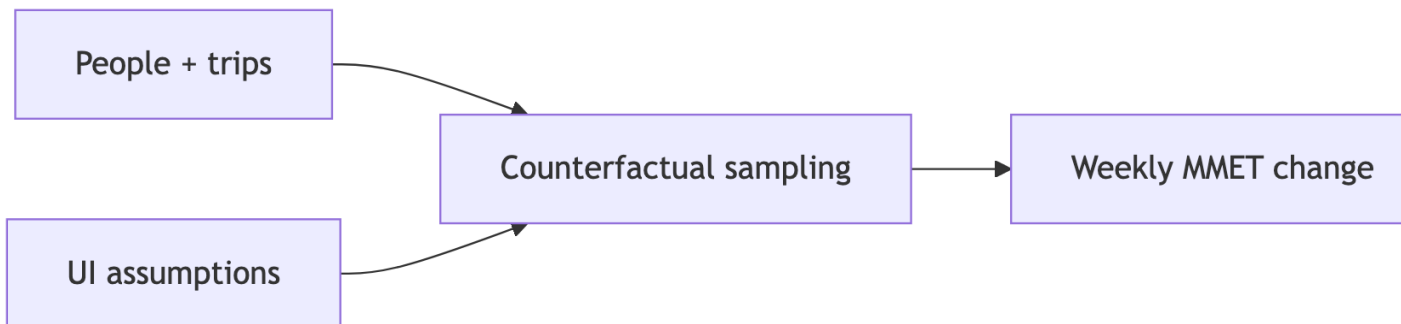
Current methodology, assumptions, and validation priorities

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What the model does

1. Selects a reference synthetic population and its weekly travel
2. Applies active-travel assumptions to a counterfactual copy
3. Converts changed walking/cycling time to MMET exposure
4. Applies the exposure change to health trajectories
5. Reports reference minus counterfactual impacts



Reference population

- Individual rows: geography, age, sex, characteristics, weekly activity
- Trip rows: mode, purpose, distance, duration, trip weight
- People and trips linked by **census_id**
- Synthetic population represents about 5% of the source population
- Current aggregate person factor: **20**
- Counterfactual population size remains fixed

Active travel to physical activity

Weekly activity	Current intensity
Walking	2.5 MMET/hour
Cycling	5.8 MMET/hour
Other vigorous activity	7.0 MMET/hour

$$\Delta MMET_i = MMET_{i,cf} - MMET_{i,ref}$$

User-based and independent trip-based changes are kept separate to avoid double counting the same active minutes.

Physical activity to health

- MIAMA-HM supplies reference person-cycle trajectories
- MMET lookup varies by age, sex, mortality risk, cycle, and outcome
- Current **long-term** assumption applies the MMET change to every cycle
- Piecewise MMET-band slopes determine outcome changes
- Internal delta: counterfactual minus reference
- Presentation impact: reference minus counterfactual
- Current assessment period: **40 years**, excluding cycle 0

Ways users can specify change

Interface	Input	Status
Basic	Mode users	Implemented
Basic	Mode trips	Implemented
Basic	Distance/duration	Planned allocation
Basic	Mode share	Planned conversion
Advanced population	User count + age/sex + PA/sex	Implemented as sampling constraints
Advanced trips	Trip count + distance + purpose + car diversion	Partly implemented

Five-category spread controls

- Reference data provide a 5 x 2 distribution for each mode
- Slider values summarize the same data: mean + binary proportion
- Exponential tilting shifts the five-category marginal
- Requested sex/purpose proportion replaces the binary marginal
- Outer product reconstructs the displayed joint distribution
- An unchanged slider reproduces the reference bars exactly

Configured groups

- Age: 18–29, 30–39, 40–49, 50–59, 60+
- Distance: 0–2, 2–5, 5–10, 10–30, 30+ km
- PA: sedentary, low, moderate, high, very high

Sampling people

User increase

- sample existing non-users as new users
- probability weights combine sex, age, and PA targets
- sample weekly mode activity from observed current users
- shift plausible trips for selected people where trip data exist

User decrease

- sample existing users as ex-users
- set mode activity to zero
- divert their active trips, currently to car

Sampling trips

Trip increase

- 90% existing non-active utilitarian trips switch mode
- 10% induced recreational trips are added
- retain observed raw distance and duration
- distance target and car-source share weight candidate selection

Trip decrease

- active trips switch away from the mode
- travel needs remain; rows are not deleted

What “matching” currently means

- Target category shares are **relative sampling weights**, not quotas
- Sampling is without replacement for people and existing trips
- Sparse combined constraints can be relaxed and are reported
- Large samples should be more stable, but exact matching is not guaranteed
- Requested versus realized distributions still need a complete audit table

Warning

Candidate weights are not yet calibrated for category prevalence in the eligible pool. The purpose slider is parsed but not yet included in trip candidate weights.

Important current limitations

- PA target selects candidates using **pre-change** PA; final PA can differ
- “Current vs new users” input is not yet operational
- Trip weights and physical row changes are not fully reconciled
- Basic mode-share and distance/duration routes do not yet change outcomes
- Only long-term persistence is implemented
- Walking/cycling are the defensible person-level active modes
- Sampling uncertainty is not yet quantified

Validation priorities

1. Apply and verify trip-purpose targeting
2. Compare requested and realized distributions
3. Calibrate probabilities or implement near-exact quotas
4. Define PA and current/new-user semantics
5. Resolve weighted trip targets
6. Operationalize remaining basic input routes
7. Add short-term effects and uncertainty analysis
8. Validate against observed data and established appraisal tools

What early testers should inspect

- Are reference counts and distributions plausible for the location?
- Did requested user/trip changes occur?
- Were sampling constraints relaxed?
- Are induced trips and diversion shares reasonable?

- Do walking/cycling MMET changes have the expected sign and magnitude?
- Are prevented health outcomes positive for beneficial scenarios?
- Is the geography large enough for stable constrained sampling?

Terminology

Term	Meaning
Reference (ref)	Without-scheme scenario
Counterfactual (cf)	With-scheme scenario
Technical delta	cf - ref
Prevented impact	ref - cf
Baseline	Reserved for the health model's initial state/cycle

Review question

Does the current model allocate the requested active-travel change to plausible people and trips **in a way that is transparent, empirically defensible, and consistent with the question asked in the UI?**

Full methods and TODO register: `docs/methodology.qmd`